



FACULTY OF BUSINESS AND IT

MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestones 3

Due: June 9, 2026

Milestone	Description	Due Date
1	Capstone project selection/proposal	May 19
2	Literature review and data source identification	May 26
3	Data collection & Initial EDA	June 9
4	Data Preprocessing & Baseline Modeling	June 23
5	Advanced Modeling & Optimization	July 7
6	Validation & Business Impact Assessment	July 21
7	Capstone project oral presentation	July 27 & August 10
8	Capstone project report	August 10

Important Note on Pace and Deadlines

Project complexity and team progress may vary throughout the term. If your group completes a milestone before its scheduled deadline, you are encouraged to begin working on the next milestone immediately rather than waiting for the due date.

The posted deadlines represent the maximum allowable completion time for each milestone and should not be interpreted as required pacing checkpoints. Early progress is strongly encouraged to allow additional time for refinement, testing, validation, and project improvement.

Important Note on Project Variability and EDA Scope

The depth and complexity of exploratory data analysis (EDA) may vary significantly depending on the nature of the capstone project, dataset characteristics, and analytical objectives. While many projects will require extensive exploratory analysis to understand complex datasets, identify patterns, evaluate feature relationships, and detect data quality issues, some projects may require only limited EDA due to the structure or maturity of the available data.

Students are encouraged to apply EDA at a level appropriate to their project requirements rather than attempting unnecessary or repetitive analysis. The primary goal of Milestone 3 is to demonstrate a sufficient understanding of the dataset and confirm readiness for preprocessing and baseline modeling.

If a project team completes the required data collection and initial EDA tasks before the milestone deadline, they are strongly encouraged to begin work on Milestone 4 immediately. Early progress into preprocessing and baseline model development is recommended to allow additional time for experimentation, optimization, validation, and project refinement later in the term.

Milestone 3: Data Collection & Initial Exploratory Data Analysis (EDA)

Milestone 3 is designed to transition the capstone project from research and planning into practical data acquisition, understanding, and early-stage analytical exploration.

Students are expected to collect, organize, and examine the selected datasets identified during Milestone 2 while demonstrating the ability to perform professional exploratory data analysis (EDA). The purpose of this milestone is to validate dataset suitability, uncover patterns and relationships within the data, identify data quality issues, and establish a strong analytical foundation for future preprocessing and machine learning implementation.

Students should demonstrate graduate-level analytical reasoning through systematic exploration of the dataset(s), interpretation of findings, visualization of trends, and assessment of potential challenges that may impact modeling and business outcomes.

The data collection and EDA process should clearly explain:

- How the dataset(s) were acquired and integrated,
- What the dataset structure and characteristics reveal,
- Which patterns, distributions, and correlations exist,
- What data quality issues are present,
- Which variables appear most relevant to the business problem,
- What preprocessing challenges are anticipated,
- How the exploratory findings influence future modeling decisions.

Students are expected to build a comprehensive understanding of their data before proceeding to preprocessing and model development in Milestone 4.

1. Data Collection and Dataset Preparation

Students must collect and organize all datasets required for the project implementation.

- **Data Collection Requirements**

Students must provide:

- Finalized primary dataset,
- Backup or supplementary datasets (if applicable),
- Data acquisition process,
- Data integration procedures,
- Data storage format and organization,
- Data extraction or API collection procedures (if applicable),
- Initial dataset verification and validation.

- **Dataset Documentation**

Students should clearly document:

- Total number of observations/records,
- Number of features/variables,
- Target variable(s),
- Data types of variables,

- Structured vs unstructured data characteristics,
- Time range covered by the dataset (if applicable),
- Dataset updates or versioning information.

Students should also explain any modifications made during data collection, including:

- Merging datasets,
- Filtering records,
- Removing duplicates,
- Handling inaccessible or incomplete data sources.

2. Exploratory Data Analysis (EDA)

Students must conduct a comprehensive exploratory analysis of the dataset(s).

• EDA Objectives

The EDA should aim to:

- Understand dataset structure and variable distributions,
- Detect anomalies and inconsistencies,
- Identify relationships between variables,
- Assess target variable behavior,
- Discover trends, patterns, and business insights,
- Identify potential predictive signals,
- Evaluate dataset suitability for machine learning tasks.

• Required EDA Components

Students are expected to include:

Descriptive Statistics

Examples may include:

- Mean,
- Median,
- Standard deviation,
- Minimum and maximum values,
- Quartiles and percentiles,
- Frequency distributions.

Missing Value Analysis

Students should identify:

- Missing data percentages,
- Missing value patterns,
- Potential causes of missingness,
- Variables requiring imputation or removal.

Outlier Detection

Students should analyze:

- Extreme values,
- Anomalous records,
- Potential noise or measurement errors,
- Business significance of outliers.

Correlation and Relationship Analysis

Students should examine:

- Feature correlations,
- Relationships with target variables,
- Multicollinearity concerns,
- Dependency patterns between variables.

Class Distribution Analysis (if applicable)

Students should evaluate:

- Balanced vs imbalanced target classes,
- Rare event occurrence,
- Distribution skewness,
- Potential impact on future modeling.

Data Visualization

Students must include meaningful visualizations such as:

- Histograms,
- Box plots,
- Scatter plots,
- Correlation heatmaps,
- Pair plots,
- Bar charts,
- Time-series trend plots,
- Geographic visualizations (if applicable).

Visualizations should support interpretation and business understanding rather than serving only decorative purposes.

3. Initial Business Insights and Interpretation

Students must interpret findings from the exploratory analysis and connect them to the project objectives. This section should explain:

- Key trends discovered in the data,
- Important variables influencing outcomes,
- Unexpected observations or anomalies,
- Potential business implications,
- Early hypotheses for predictive modeling,

- Data limitations affecting business interpretation.

Students are encouraged to explain how EDA findings may influence future feature engineering, model selection, and evaluation strategies.

4. Preliminary Data Quality Assessment

Students must evaluate the overall quality and usability of the dataset(s).

The assessment should discuss:

- Data completeness and consistency across records,
- Duplicate, noise and inconsistencies,
- Formatting issues,
- Data integrity concerns and reliability of data sources,
- Potential bias within the dataset.

Students should identify any major data preparation challenges expected in Milestone 4.

5. Preliminary Preprocessing Plan

Students must propose a preliminary plan for preparing the data for machine learning and analytical modeling.

The preprocessing plan may include:

- Missing value handling strategies,
- Encoding categorical variables,
- Feature scaling or normalization,
- Outlier treatment methods,
- Feature selection approaches,
- Dimensionality reduction techniques,
- Text preprocessing (for NLP projects),
- Time-series transformation methods,
- Data balancing techniques and data splitting strategies.

Students should explain why the proposed preprocessing methods are appropriate for their dataset and analytical goals.

6. Initial Modeling Readiness Assessment

Students must evaluate whether the dataset is ready for baseline modeling in Milestone 4.

The readiness assessment should discuss:

- Suitability of features for modeling,
- Expected modeling challenges,
- Computational considerations,
- Dataset size adequacy,
- Potential overfitting concerns,
- Feature engineering opportunities.

Students should also identify any remaining risks that could impact project implementation.

Milestone 3 Deliverables

Students must submit one professionally formatted document (PDF or Word) containing the following sections:

1. Title Page

The title page should include:

- Project Title
- Student Group Name
- Full Names of Team Members

2. Updated Executive Summary (150–250 words)

The updated summary should briefly describe:

- Data collection progress,
- Dataset characteristics,
- Major EDA findings,
- Initial business insights,
- Data quality observations,
- Planned preprocessing strategies,
- Expected next steps for modeling.

3. Data Collection and Dataset Description

Students must include:

- Finalized dataset source(s),
- Data acquisition procedures,
- Dataset structure and characteristics,
- Variable descriptions,
- Data storage formats,
- Data integration procedures,
- Dataset limitations.

4. Exploratory Data Analysis (EDA)

This section must include:

- **Statistical Summary**
 - Descriptive statistics,
 - Distribution analysis,
 - Correlation analysis,
 - Missing value analysis,
 - Outlier analysis.

- **Data Visualization**

Students must provide:

- Relevant charts and graphs,
- Interpretation of visual findings,
- Business-oriented observations.

- **Analytical Interpretation**

Students should explain:

- Significant discoveries,
- Relationships between variables,
- Patterns affecting the business problem,
- Early predictive insights.

5. Data Quality and Preprocessing Assessment

Students must discuss:

- Data quality concerns,
- Missing data handling plans,
- Outlier treatment approaches,
- Planned preprocessing techniques,
- Anticipated transformation requirements.

6. Modeling Readiness Assessment

Students should discuss:

- Suitability for machine learning,
- Potential modeling limitations,
- Computational feasibility,
- Risks and mitigation strategies,
- Preparation status for baseline modeling.

7. References (IEEE Style)

Students must use proper IEEE citation formatting consistently throughout the document.

Example References

[1] J. Smith and A. Kumar, "Explainable ensemble learning for customer churn prediction," *IEEE Access*, vol. 13, pp. 11234–11249, 2025.

[2] R. Chen et al., "Comparative analysis of deep learning and traditional machine learning methods for financial forecasting," *Expert Systems with Applications*, vol. 245, 2026.

[3] L. Johnson and M. Patel, "A hybrid NLP framework for social media sentiment analytics," *Proceedings of the IEEE International Conference on Data Science and Advanced Analytics*, 2024.

Submission Guideline

Please submit one Data Collection and Initial EDA report per group through Canvas no later than **June 9, 2026, at 11:59 PM**.